

Discrete-Time Current Control of Voltage-Fed Three-Phase PWM Inverters

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Abstract— The discrete-time current control of three-phase voltage-fed pulsewidth modulation (PWM) inverters is discussed, with emphasis on important practical aspects. The inverter feeds a balanced three-phase load of R-L impedances in series with back-emf's. The inverter-load system is modeled by using space vectors. The practical difficulties of the predictive control method are pointed out. In particular, effects of the delayed application of the manipulated voltage caused by controller computing time are investigated. A modified control law that eliminates the undesirable effects of the computing delay is introduced. Also, prediction of current reference is utilized, which would enable operation with any reference function. An estimation and prediction procedure for the back emf is also introduced. Steady-state errors in the load current due to these predictions are calculated for sinusoidal operation. Simulation results for the predictive control and the modified control laws are presented and discussed.

I. INTRODUCTION

IN the indirect vector control of induction machines, torque and flux components of the stator current in a stationary two-axis reference frame are generated by the control system as manipulated variables. The pulsewidth modulation (PWM) inverter driving the machine is then required to excite the machine with these current commands. In recent years, there has been a tendency to implement the current control loop of a vector control scheme using voltage-fed inverters (VFI). Various techniques have been devised by many researchers for controlling the output current of a PWM voltage-fed inverter. Particularly, hysteresis control has found widespread application because of the simplicity of its implementation [1], [2]. However, this method has the inherent drawback that a phase current does not respond to only the respective pole voltage of a three-phase half-bridge inverter feeding a load with no neutral connection. This causes excursions of a phase current out of the hysteresis band, and the actual current error can become double the hysteresis width [2]. Furthermore, the switching frequency of an inverter in hysteresis control techniques is strongly dependent on operating conditions as well as on load parameters. Also, the switching frequency can have a wide variation within one output cycle [2]. Various techniques have been proposed in the literature to overcome the above-mentioned disadvantages of this approach. In the adaptive hysteresis-band technique [3], the band is programmed as a function of load and supply parameters in order to keep the switching frequency nearly constant. In the ramp comparison

control method [2], current errors are compared with a triangular carrier wave and the resulting PWM signals are used to control the inverter switches. In other words, current errors are used as the modulating waves in the well-known sine-triangle PWM method. The switching frequency of the inverter is almost constant around the carrier frequency. This approach has the difficulty that multiple crossings of the carrier wave by the current error waves occur when the time rate of change of error currents exceeds the slope of the triangular carrier. There is also magnitude and phase errors in the controlled phase currents.

The three-dependent hysteresis controller approaches make use of the space vector concept to coordinate the switches of the three phases and utilize zero voltage vectors to reduce the switching frequency [4]–[6].

On the other hand, feedback control of inverter current in discrete time has the obvious advantage that the switching frequency can be kept constant under all operating conditions. The constant switching frequency predictive control method [2], [7] is basically a discrete-time deadbeat control technique based on discretization of the differential equations describing the inverter-load system. In the minimum switching frequency predictive control method [12], the current trajectory that takes the longest time to hit the hysteresis boundary and the inverter voltage vector that produces this trajectory are predicted.

Current control techniques have also been devised for three-phase PWM ac/dc converters [8]–[10]. In the predicted current control with fixed switching frequency (PCFF) [9], [10] the duty cycle for each converter leg is obtained, which forces the corresponding ac phase current to follow the current reference. In ac/dc converters, in addition to achieving a desired power factor (usually unity), another objective is to regulate the output dc voltage. Hence, the current reference is partly derived from a feedback loop for output voltage control and partly from the input ac voltage for power factor control.

In this paper, various aspects of the constant switching frequency predictive control method are discussed. In particular, effects of the time delay between control sampling and the application of the manipulated voltage vector to the system are considered. A modified control law that eliminates this time delay is proposed. Both the predictive and the modified control laws require that the current reference and back-emf vectors be predicted from previously acquired or estimated values. Hence, one-step- and two-step-ahead extrapolations are utilized. Steady-state errors in the controlled current for sinusoidal operation due to the current reference and back-emf

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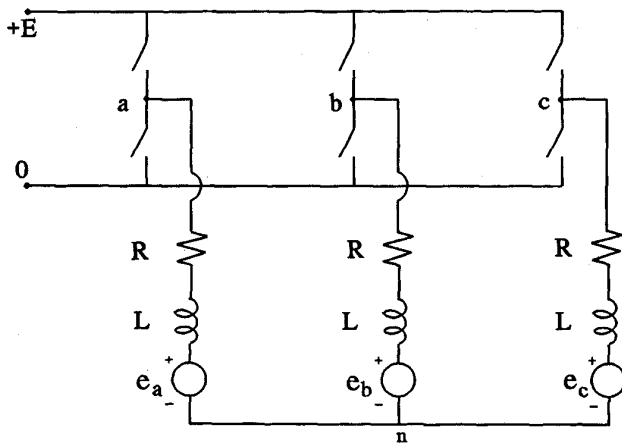


Fig. 1. A three-phase half-bridge inverter with R-L load and back-emf's.

predictions are calculated. Computer simulations for various steady-state and transient operations are performed.

II. HALF-BRIDGE INVERTER MODELING

Fig. 1 depicts a three-phase half-bridge inverter fed from a single dc voltage source and feeding a balanced Y-connected load of R-L impedances in series with back emf's. If the neutral point of the load is not connected to the midpoint of a split dc supply, the phase voltages of the load are related to the pole voltages of the inverter by

$$\begin{bmatrix} v_{an} \\ v_{bn} \\ v_{cn} \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} \begin{bmatrix} v_{ao} \\ v_{bo} \\ v_{co} \end{bmatrix}.$$

The phase voltages can be represented by the following space vector

$$\mathbf{V} = \frac{2}{3} \{v_{an} + \bar{\mathbf{u}}v_{bn} + \bar{\mathbf{u}}^2v_{cn}\}; \quad \bar{\mathbf{u}} = e^{j2\pi/3}. \quad (1)$$

Corresponding to each switching state of the inverter, eight voltage vectors will be generated, as shown in Fig. 2, which can be expressed as

$$\begin{aligned} \mathbf{V}_m &= \frac{2}{3}Ee^{j(m-1)\pi/3} \quad m = 1, \dots, 6 \\ \mathbf{V}_7 &= \mathbf{V}_8 = \mathbf{0}. \end{aligned} \quad (2)$$

The zero voltage vectors correspond to the switching states of the inverter in which either the upper or the lower switches are closed.

For each phase of the load, the following differential equation can be written

$$v_{pn} = Ri_p + L \frac{di_p}{dt} + e_p \quad p = a, b, c. \quad (3)$$

Equation (3) can be combined to obtain the vector equation

$$\mathbf{V} = R\mathbf{I} + L \frac{d\mathbf{I}}{dt} + \mathbf{E}. \quad (4)$$

In (4)

$$\begin{aligned} \mathbf{I} &= \frac{2}{3}(i_a + \bar{\mathbf{u}}i_b + \bar{\mathbf{u}}^2i_c) \\ \mathbf{E} &= \frac{2}{3}(e_a + \bar{\mathbf{u}}e_b + \bar{\mathbf{u}}^2e_c). \end{aligned}$$

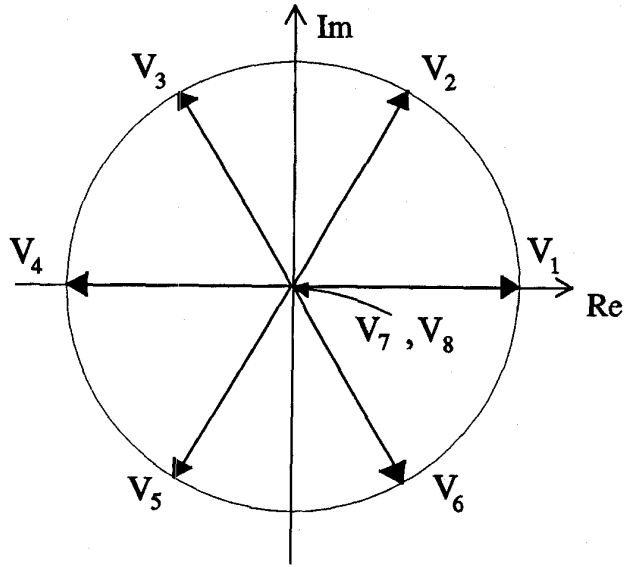


Fig. 2. Voltage vectors of a three-phase half-bridge inverter.

In (4) it will be assumed that $\mathbf{V}$ represents any voltage vector and not one of the vectors generated by the inverter. If it is assumed that $\mathbf{V}$ and $\mathbf{E}$ are constant between sampling instants kT and $(k+1)T$, then (4) can be discretized as follows:

$$\begin{aligned} \mathbf{I}[(k+1)T] &= e^{\alpha T} \mathbf{I}[kT] + \int_{kT}^{(k+1)T} e^{\alpha[(k+1)T-t']} dt' \\ &\quad \cdot \frac{1}{L}(\mathbf{V} - \mathbf{E}) \end{aligned} \quad (5)$$

where $\alpha = -R/L$. Equation (5) reduces to the following discrete-time model for the inverter-load system

$$\frac{1}{b} \mathbf{I}(k+1) - \frac{a}{b} \mathbf{I}(k) = \mathbf{V}(k) - \mathbf{E}(k) \quad (6)$$

where

$$a = e^{-T/\tau}; \quad b = \frac{1}{R}(1 - e^{-T/\tau}); \quad \tau = \frac{L}{R}.$$

III. DISCRETE-TIME CURRENT CONTROL

For the discrete-time control of the system in (6), the current error vector is defined as

$$\Delta \mathbf{I}(k) = \mathbf{I}^*(k) - \mathbf{I}(k) \quad (7)$$

where $\mathbf{I}^*(k)$ is the sampled reference current vector. Substitution of (7) in (6) gives

$$\begin{aligned} \{\mathbf{I}^*(k+1) - a\mathbf{I}^*(k)\} - \Delta \mathbf{I}(k+1) + a\Delta \mathbf{I}(k) \\ = b\{\mathbf{V}(k) - \mathbf{E}(k)\}. \end{aligned} \quad (8)$$

Let the voltage vector $\mathbf{V}(k)$ in (8) be written as

$$\mathbf{V}(k) = \mathbf{V}^*(k) + \Delta \mathbf{V}(k) \quad (9)$$

where $\mathbf{V}^*(k)$ is defined as the reference for the voltage vector. If this voltage reference is expressed as

$$\mathbf{V}^*(k) = \mathbf{E}(k) + \frac{1}{b} \{\mathbf{I}^*(k+1) - a\mathbf{I}^*(k)\} \quad (10)$$

then (8) reduces to

$$\Delta I(k+1) = a\Delta I(k) - b\Delta V(k). \quad (11)$$

Now, with the following state feedback

$$\Delta V(k) = K_0 \Delta I(k) \quad (12)$$

(11) becomes

$$\Delta I(k+1) = (a - bK_0)\Delta I(k). \quad (13)$$

Hence, deadbeat control in one control period is obtained if

$$K_0 = \frac{a}{b}. \quad (14)$$

With the gain K_0 as in (14) and with an arbitrary current error $\Delta I(k)$ at step k , $\Delta I(k+1) = 0$, assuming that the input magnitude $|V(k)|$ is not bounded.

In (10), calculation of the voltage reference vector at control step k requires the knowledge of current reference vectors at steps k and $(k+1)$. However, unless the reference current is a specified function of time, then $I^*(k+1)$ is not known. Therefore, it must be predicted using previously acquired values. Denoting by $I_p^*(k+1)$ the predicted current reference, the current error with the control law in (12) is obtained as

$$\Delta I(k+1) = I^*(k+1) - I_p^*(k+1). \quad (15)$$

Equation (15) implies that the steady-state current error is equal to the prediction error of its reference. Hence, an efficient prediction method must be used, as described in the following section.

In addition to the current reference, the back-emf vector is also unknown in (10). Therefore, it must be estimated from measured quantities, which are the current vectors. Now, suppose that $E(k)$ is predicted using previously estimated values, and denoted by $E_p(k)$. Subsequently, the voltage vector $V(k)$ is calculated and applied to the load in the k th sampling interval ($kT \leq t \leq (k+1)T$). Then, the current vector $I(k+1)$ is measured at the instant $(k+1)T$. The back-emf can then be estimated from (6). Thus

$$E_e(k) = V(k) - \frac{1}{b}(I(k+1) - aI(k)) \quad (16)$$

where subscript e denotes estimated value. This estimated value of emf together with previously calculated estimates can be used to predict the emf $E_p(k+1)$ in the $(k+1)$ st sampling interval, which is then used to calculate $V^*(k+1)$ according to (10). Since the voltage vector would be implemented by employing a PWM method, the estimated emf would differ from its actual value due to this PWM approximation. This estimation and prediction process for the back-emf vector gives rise to an error in the controlled current vector. In order to find out this error, let

$$P^*(k) = \frac{1}{b}\{I^*(k+1) - aI^*(k)\}. \quad (17)$$

In (17) assume that the current reference vector is known. Then

$$V(k) = E_p(k) + P^*(k) + \Delta V(k). \quad (18)$$

Substitution of (18) in (8) gives

$$\Delta I(k+1) = b[E(k) - E_p(k)]. \quad (19)$$

Evidently, the current errors in (19) and (15) do not account for the errors due to the PWM approximation of $V(k)$.

IV. PREDICTION OF CURRENT REFERENCE AND BACK-EMF

One-step-ahead prediction of the current reference can be obtained from the Lagrange interpolation formula [11] by making use of the fact that the sampling period is constant. The n th order prediction is the following

$$I_p^*(k+1) = \sum_{l=0}^n (-1)^{n-l} \binom{n+1}{l} I^*(k+l-n). \quad (20)$$

For $n = 1$, one obtains from (20) the linear prediction

$$I_p^*(k+1) = 2I^*(k) - I^*(k-1) \quad (21)$$

and for $n = 2$, the quadratic prediction

$$I^*(k+1) = 3I^*(k) - 3I^*(k-1) + I^*(k-2). \quad (22)$$

For sinusoidal $I^*(k)$

$$I^*(k) = I_o e^{jk\omega T} = I_o q^k \quad (23)$$

where $q = e^{j\theta}$ and $\theta = \omega T$. The prediction error can be evaluated as

$$I_p^*(k+1) - I^*(k+1) = -I_o q^{k-n}(q-1)^{n+1}. \quad (24)$$

Therefore, the current error in (15) has the normalized magnitude given by

$$\left| \frac{\Delta I(k+1)}{I_o} \right| = |(q-1)|^{n+1} = 2^{n+1} \sin^{(n+1)}\left(\frac{\theta}{2}\right). \quad (25)$$

For a sampling frequency of $f_s = 2.0$ kHz and for an operating frequency of $f = 50$ Hz, the error of linear prediction is 2.46% and that of quadratic prediction is 0.386%. At $f = 100$ Hz, these errors increase to 9.8% and 3.06%, respectively. Therefore, if the operating frequency range is wide, at least quadratic prediction must be used.

V. EFFECT OF COMPUTING DELAY

In a practical implementation of the current controller described before, there would be a time delay between acquisition of the current error at the sampling instant and the application of the manipulated voltage vector to the system. This delay is caused by the control calculations comprising predictions of the current reference and emf vectors, computation of the voltage reference and correction vectors (in (10) and (12)), and the PWM approximation of the voltage vector $V(k)$ (see Section VII). It is essential to investigate what effect this delay would have on the steady-state and dynamic performance of the control system. In the following analysis, PWM approximation of the voltage vector will not be taken into account to simplify the analysis. In addition, predictions of reference and emf vectors will not be considered.

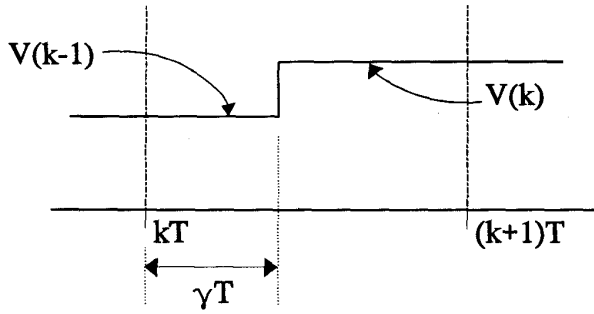


Fig. 3. Delayed application of the voltage vector in a sampling interval.

Referring to Fig. 3, the current vectors at the instants $(k + \gamma)T$ and $(k + 1)T$ are given by

$$I[(k + \gamma)T] = a_1 I(kT) + b_1 [V(k-1) - E(k)] \quad (26)$$

$$I[(k + 1)T] = a_2 I[(k + \gamma)T] + b_2 [V(k) - E(k)] \quad (27)$$

where

$$a_1 = e^{-\gamma T/\tau} \quad b_1 = \frac{1 - a_1}{R} \quad (28)$$

$$a_2 = e^{-(1-\gamma)T/\tau} \quad b_2 = \frac{1 - a_2}{R}. \quad (29)$$

Substitution of (26) in (27) gives

$$I(k + 1) = aI(k) + (b - b_2)V(k-1) + b_2V(k) - bE(k). \quad (30)$$

Now, if (30) is written in terms of current errors, also making use of (9), (10), and (12), we obtain the following difference equation for the system

$$\begin{aligned} \Delta I(k + 1) - (a - b_2 K_0) \Delta I(k) + b_3 K_0 \Delta I(k - 1) \\ = \frac{b_3}{b} \{I^*(k + 1) - (1 + a)I^*(k) + aI^*(k - 1)\} \\ + b_3 \{E(k) - E(k - 1)\} \end{aligned} \quad (31)$$

where $b_3 = b - b_2$. For small T/τ , the ratio (b_3/b) can be approximated by γ and then the transfer function relating $\Delta I(z)$ to $I^*(z)$ can be written as

$$H_1(z) = \frac{\gamma[z^2 - (1 + a)z + a]}{z^2 - a\gamma z + a\gamma} \quad (32)$$

and the transfer function relating $\Delta I(z)$ to $E(z)$ is

$$H_2(z) = \frac{\gamma b(z - 1)}{z^2 - a\gamma z + a\gamma}. \quad (33)$$

Since the steady-state response to a complex sinusoid $e^{jk\Theta}$ is $H(e^{j\Theta})e^{jk\Theta}$, where $H(z)$ is the transfer function, the current error vector in the steady state is

$$\begin{aligned} \Delta I_{ss}(k) = \gamma I_0 e^{jk\Theta} \left[\frac{q^2 - (1 + a)q + a}{q^2 - a\gamma q + a\gamma} \right] \\ + \gamma b E_0 e^{j\phi} e^{jk\Theta} \left[\frac{q - 1}{q^2 - a\gamma q + a\gamma} \right]. \end{aligned} \quad (34)$$

In (34) the back-emf vector is taken as

$$E(k) = E_0 e^{j(k\Theta + \phi)}. \quad (35)$$

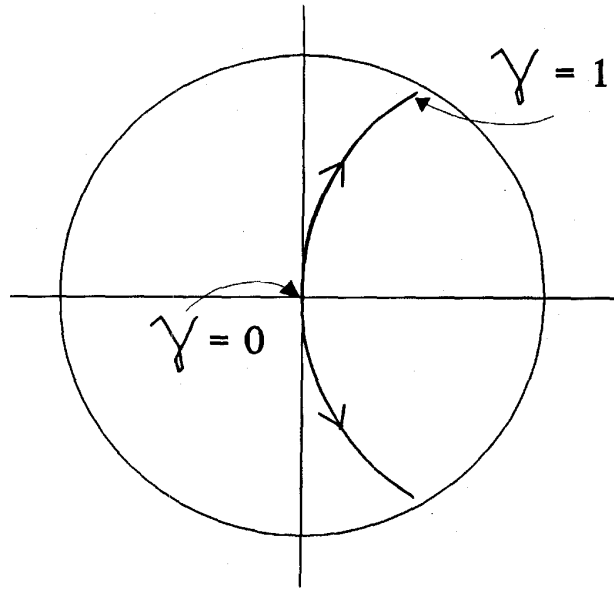


Fig. 4. Loci of closed-loop poles with delayed application of voltage.

The magnitude of the first term normalized to the current amplitude I_0 can be verified to be negligibly small for reasonable values of γ and Θ . However, the amplitude of the second term can be shown to be given approximately by

$$|\Delta I|_2 \simeq \gamma \left(\frac{E_0}{\omega L} \right) \Theta^2 \quad (36)$$

which can attain large values if the emf amplitude and the sampling angle are large.

The characteristic equation of the system in (32) is

$$z^2 - a\gamma z + a\gamma = 0. \quad (37)$$

Therefore, the poles of the system are

$$z_{1,2} = \frac{1}{2}a\gamma \pm j \frac{1}{2}\sqrt{4a\gamma - a^2\gamma^2} \quad (38)$$

which are complex for $0 < \gamma < 1$. Fig. 4 shows the loci of the poles. Although the system is asymptotically stable for $0 < \gamma < 1$, the response is highly oscillatory as the argument of a pole is large (approximately in the range 60° – 90°). For instance, for $a = 0.9$ and $\gamma = 1$, the poles are $z_{1,2} = 0.4524 \pm j0.8367 = 0.9512e^{\pm j61.6^\circ}$. With these poles, the settling time of the response to a step change in reference (to 5% of reference amplitude) would be 60 sampling periods, and the oscillation period would be 52.6° .

VI. MODIFIED FEEDBACK LAW

The odd effects of computing delay can be eliminated by modifying the feedback law in such a way that the current feedback at the instant of sampling is not required. The proposed controller is the following:

$$\Delta V(k) = \sum_{m=1}^N K_m \Delta I(k - m). \quad (39)$$

Equation (39) implies that as soon as the last current error $\Delta I(k-1)$ at instant $(k-1)T$ is acquired, the voltage vector to be applied at instant kT can be calculated. Thus, the time made available for computations is the full sampling period T . However, a disadvantage is that since computation of $V(k)$ starts at $(k-1)T$, the current references $I^*(k)$ and $I^*(k+1)$ must be predicted simultaneously. Prediction of $I^*(k)$ can be performed as described before, but $I^*(k+1)$ has to be predicted by using a two-step-ahead extrapolation. Hence, the quadratic prediction of $I^*(k+1)$ is

$$I_p^*(k+1) = 6I^*(k-1) - 8I^*(k-2) + 3I^*(k-3). \quad (40)$$

Similarly, the emf vector $E_p(k)$ required by the calculation of $V(k)$ must also be predicted by a two-step-ahead formula, since the estimated emf $E_e(k-1)$ cannot be calculated until the current vector $I(k)$ is measured at instant kT . Hence, quadratic prediction of emf

$$E_p(k) = 6E_e(k-2) - 8E_e(k-3) + 3E_e(k-4). \quad (41)$$

With the feedback law as in (39), the discrete-time equation describing the system is

$$\begin{aligned} \Delta I(k+1) - a\Delta I(k) + \sum_{m=1}^N bK_m \Delta I(k-m) \\ = [I^*(k+1) - I_p^*(k+1)] - a[I^*(k) - I_p^*(k)]. \end{aligned} \quad (42)$$

In (42) the estimation and prediction error of the emf is omitted for simplicity. The characteristic equation of the system is

$$z^{N+1} - az^N + \sum_{m=1}^N bK_m z^{N-m} = 0. \quad (43)$$

In (43) the $(N+1)$ roots satisfy

$$\sum_{i=1}^{N+1} z_i = a. \quad (44)$$

Therefore, the gains can be chosen to place all the roots at

$$z_p = \frac{a}{N+1}. \quad (45)$$

Hence, the characteristic equation would be

$$(z - z_p)^{N+1} = 0. \quad (46)$$

From the binomial expansion of (46) the gains are obtained as

$$K_m = \frac{1}{b}(-1)^{m+1} \binom{N+1}{m+1} \left(\frac{a}{N+1}\right)^{m+1}. \quad (47)$$

Equation (45) implies that the poles of the system can be placed as near origin as desired by increasing N . The normalized steady-state current error magnitude with the one-step and two-step-ahead quadratic predictions of current references in (22) and (40) is

$$\begin{aligned} \left| \frac{\Delta I(k)}{I_0} \right|_{ss} \\ = \left| \frac{(q-1)[q^3 + (1-a)q^2 - (5-2a)q + (3-a)]}{(q-z_p)^{N+1}} \right| \end{aligned} \quad (48)$$

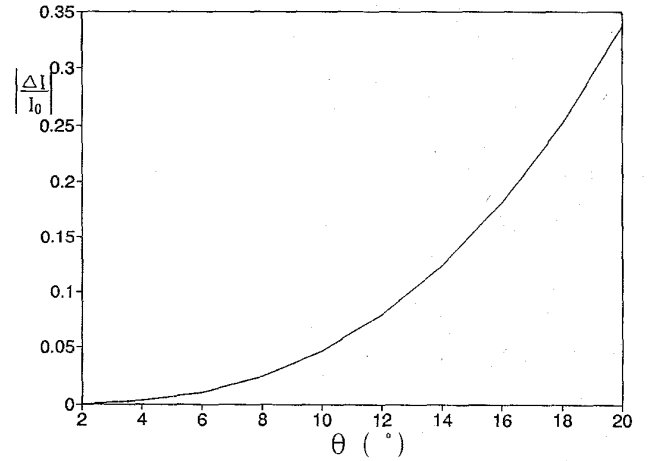


Fig. 5. Steady-state normalized current error due to current reference prediction.

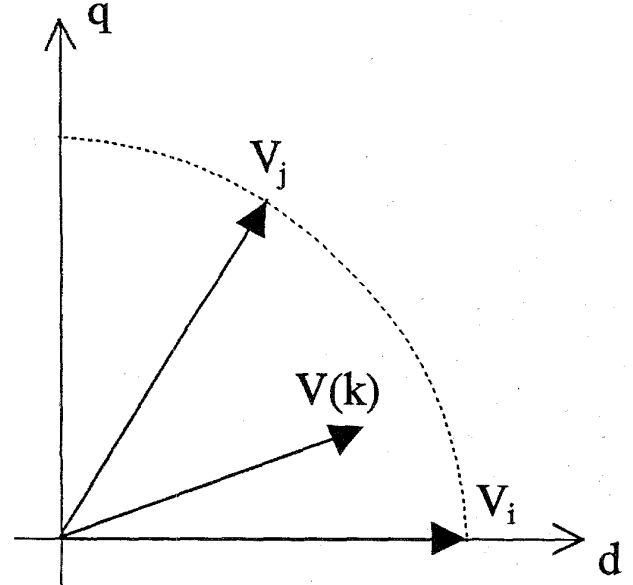


Fig. 6. Voltage vector V in sector between V_i and V_j .

where (40) has been used for $I_p^*(k+1)$. In Fig. 5 this error is plotted as a function of the sampling angle Θ for $T/\tau = 0.1$ and $N = 2$. This plot shows that the error very rapidly increases with Θ , and becomes unacceptably large for large values of Θ .

VII. PWM APPROXIMATION OF VOLTAGE

The voltage vector $V(k)$ computed by the control algorithm is an arbitrary complex quantity and evidently cannot be generated by the inverter. Hence, it must be approximated by the available voltage vectors $V_m, m = 1, \dots, 8$. One approach is to transform the d - and q -axis components, $V_d(k)$ and $V_q(k)$ to three-phase and generate the switching functions by symmetrical sampling using a triangular carrier synchronized with the control sampling. This approach has the advantage that the PWM switching functions can be generated by external hardware, thus reducing computation time. A second approach

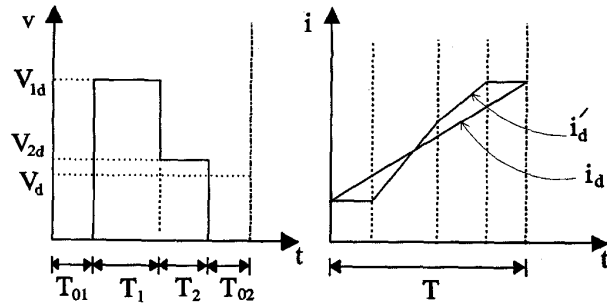


Fig. 7. Sequence of voltage vectors in sampling interval (only d -axis quantities shown, $i = 1$).

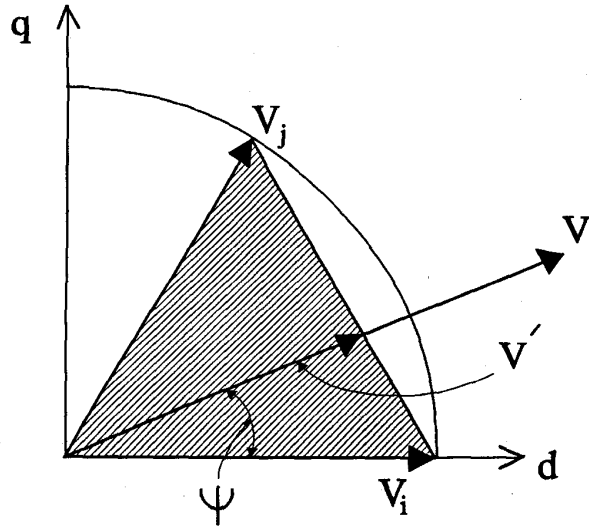


Fig. 8. Region of synthesizable voltage vectors (shaded area).

is to directly calculate the switching functions by the space vector method.

In Fig. 6 the voltage vector $V(k)$ is shown in the sector defined by V_i and V_j , where the latter are two consecutive nonzero voltage vectors of the inverter ($j = i + 1$). The vector $V(k)$ can be synthesized as the sum of the inverter voltage vectors. In the d - and q -axes

$$TV_d = T_1 V_{id} + T_2 V_{jd} \quad (49)$$

$$TV_q = T_1 V_{iq} + T_2 V_{jq} \quad (50)$$

Equations (49) and (50) imply that with a purely inductive load the same d - and q -axes currents would be obtained at the end of a sampling interval when V_i and V_j are applied for durations T_1 and T_2 instead of $V(k)$.

Solving for T_1 and T_2

$$T_1 = \frac{1}{\Delta_{ij}} [V_{jq} V_d - V_{jd} V_q] T \quad (51)$$

$$T_2 = \frac{1}{\Delta_{ij}} [V_{id} V_q - V_{iq} V_d] T \quad (52)$$

where

$$\Delta_{ij} = V_{jq} V_{id} - V_{jd} V_{iq}.$$

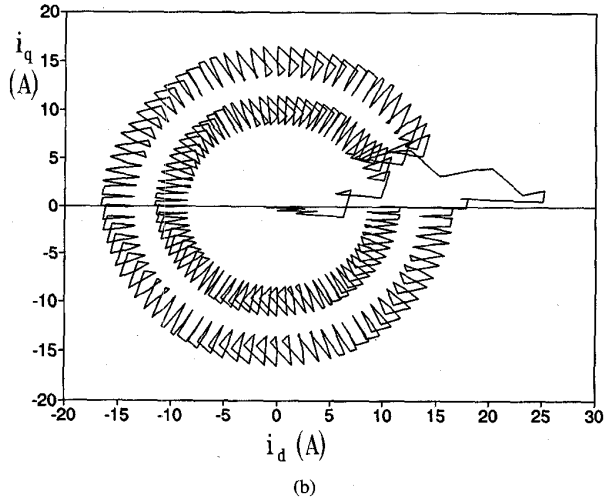
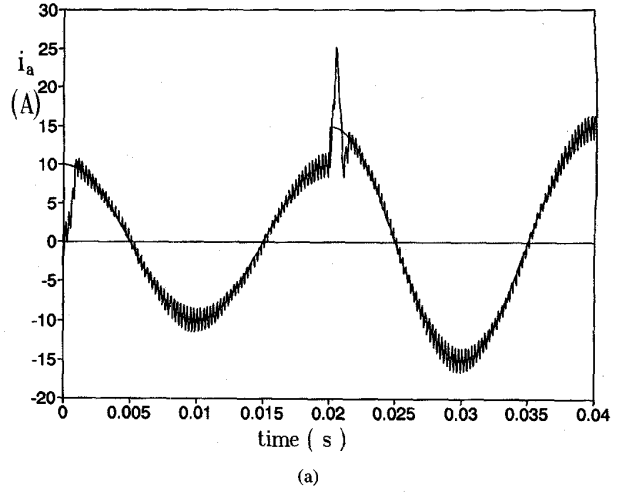


Fig. 9. (a) Response of phase A current with the predictive controller. (b) Current vector for (a).

The sum of the durations T_1 and T_2 is not in general equal to T . This means that a zero vector must also be used during a sampling interval if $T_1 + T_2 < T$. The three inverter vectors ($0, V_i, V_j$) can be applied in such a way as to minimize the following performance criterion (in the k th sampling interval)

$$J = \int_0^T |I(t) - I'(t)|^2 dt \quad (53)$$

where $I(t)$ is the load current vector created by V and $I'(t)$ is the current vector created by the inverter voltage vectors. Minimization of J would require dividing the duration of the zero vector (T_0) into two parts. Finding the optimum division of T_0 to minimize J would produce a result requiring excessive computation time in a practical application. A simple approximation can be made if it is noted that when one of the vectors V_i or V_j is absent, minimum of J occurs at $T_{01} = T_0/2$ (see Fig. 7).

The case $T_1 + T_2 > T$ means that $V(k)$ cannot be synthesized with the available voltage vectors in the sampling interval T . In this case, the voltage vector $V(k)$ must be limited in

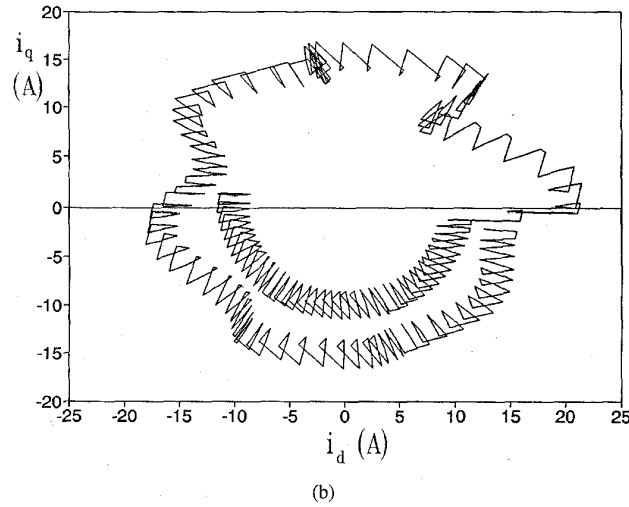
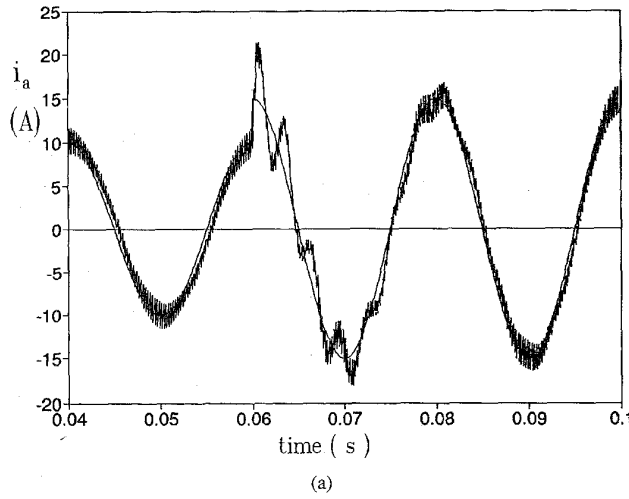


Fig. 10. (a) Response of phase A current to a step change in reference, $\gamma = 1$. (b) Current vector for (a).

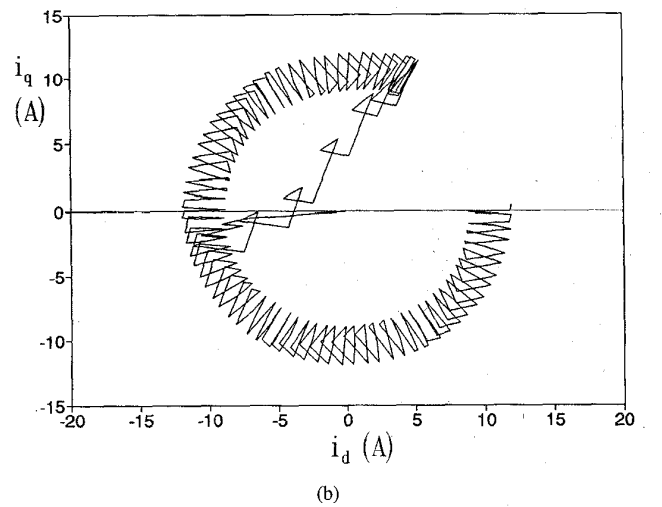
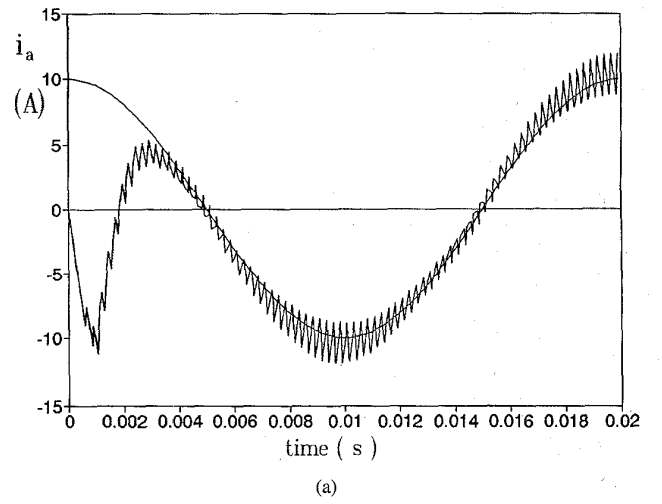


Fig. 11. (a) Response of phase A current with the modified controller, $f = 50$ Hz. (b) Current vector for (a).

magnitude. The sum of the durations T_1 and T_2 can be written as

$$T_1 + T_2 = \frac{\text{Im}[(V_j - V_i)\bar{V}]}{\text{Im}[V_j\bar{V}_i]}T \quad (54)$$

where $\bar{}$ denotes complex conjugate. Equation (54) can also be written as

$$T_1 + T_2 = \sqrt{3} \left(\frac{V}{E} \right) \sin \left(\frac{2\pi}{3} - \psi \right) T \quad (55)$$

where ψ is defined in Fig. 8. The condition that $V(k)$ can be synthesized is then

$$V \sin \left(\frac{2\pi}{3} - \psi \right) < \frac{1}{\sqrt{3}} E. \quad (56)$$

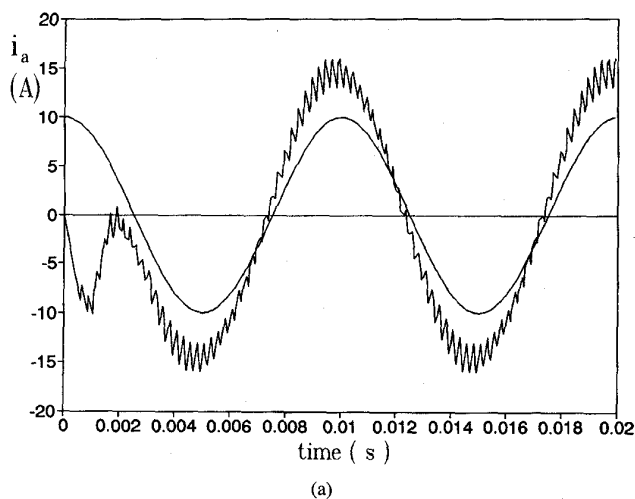
Equation (56) defines a region of synthesizable voltage vectors in the sector defined by V_i and V_j , as shown in Fig. 8. Using (55) it can be shown that the scaling of the durations T_1 and

T_2 given in (57) will scale the magnitude of the vector V to a point (V') on the boundary in Fig. 8

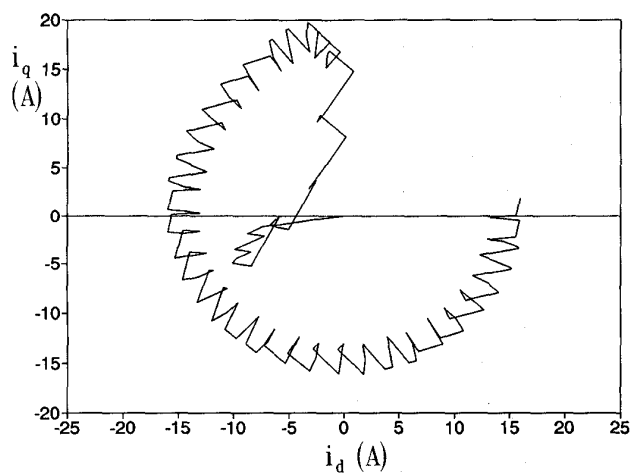
$$T'_1 = \frac{T_1}{T_1 + T_2} T \quad T'_2 = \frac{T_2}{T_1 + T_2} T. \quad (57)$$

VIII. COMPUTER SIMULATIONS

The proposed current control scheme has been simulated to investigate its performance under various operating conditions. The deadbeat control law in (12) has also been simulated for comparison. In the simulations, load parameters have been chosen as $R = 1 \Omega$, $L = 5$ mH, and $T = 0.5$ ms, $E = 450$ V. The back-emf in each phase of the load is taken to be in phase with the current and the amplitude of back-emf is taken as 80 V. The sequence of voltage vectors has been applied twice in each sampling period to reduce current ripple. When the sequence is applied once, it was observed that the ripple is excessive, as can be expected since the back-emf amplitude is small and inverter modulation level is low.



(a)

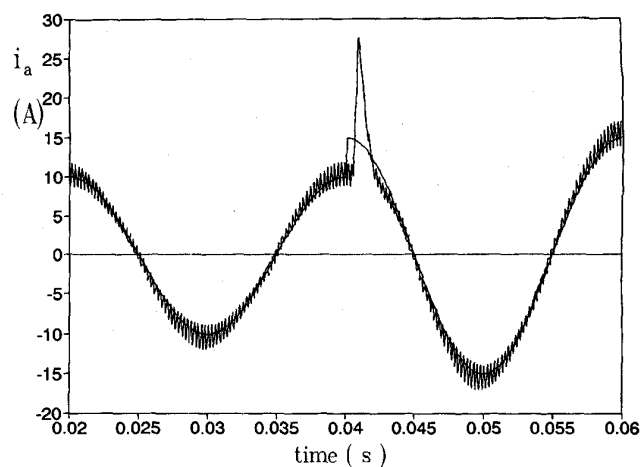


(b)

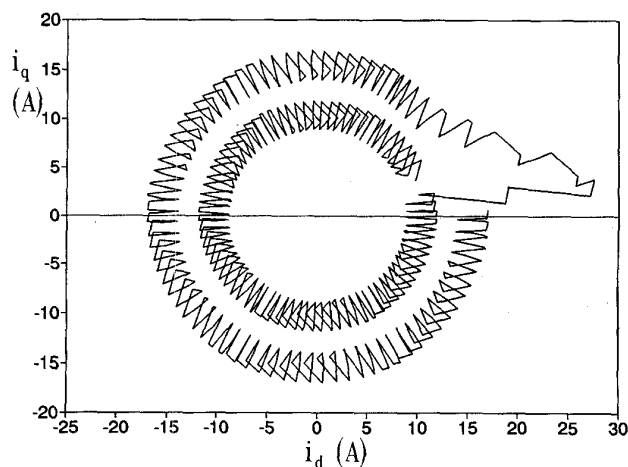
Fig. 12. (a) Response of phase A current with the modified controller, $f = 100$ Hz. (b) Current vector for (a).

Fig. 9 shows the response of phase A current and the corresponding current vector when the deadbeat control law in (12) is used. This figure also displays the response to a step change in reference current from 10 A to 15 A applied at the start of the second cycle. The current spike is caused by the reference prediction, which generates an unrealistic prediction when a step change in the reference current occurs.

Simulation of the delayed application of voltage to the load is quite difficult. But the extreme case $\gamma = 1$, which corresponds to a delay equal to one sampling period, can be easily simulated. Fig. 10 shows the response of phase A current to a step change in reference current from 10 A to 15 A amplitude with $\gamma = 1$. The oscillation period of the current error can be measured from Fig. 10 (a) to be approximately 50° , which agrees well with the theoretical value predicted before. Also, it can be observed that the settling time is approximately one and a half periods of $i_a(t)$, which is equivalent to 60 sampling periods. This also agrees with the theoretical value of 60 periods. The steady-



(a)



(b)

Fig. 13. (a) Response of phase A current to a step change in reference with modified controller. (b) Current vector for (a).

state error predicted in (34) is also visible in these responses. Fig. 11 shows the response of phase A current and the current vector at an operating frequency of 50 Hz when the modified controller in (39) is used with $N = 2$. The steady-state error is noticeable in the response of i_a . The steady-state current error becomes clearly visible at higher operating frequencies, as can be observed in Fig. 12, which corresponds to $f = 100$ Hz. The current error magnitude here is around 40%. The theoretically predicted error magnitude for this case ($\Theta = 18^\circ$) in Fig. 5 is 26%. The difference between the theoretical and the simulation results is attributed to the contribution of the emf prediction to the error in the simulation. These results show that a higher order prediction is essential to keep the steady-state current error within acceptable limits. The sampling angle Θ at the highest operating frequency may also be limited to reduce the error. However, the smallest Θ is determined by computation requirement of the controlling processor or the maximum switching frequency of the inverter.

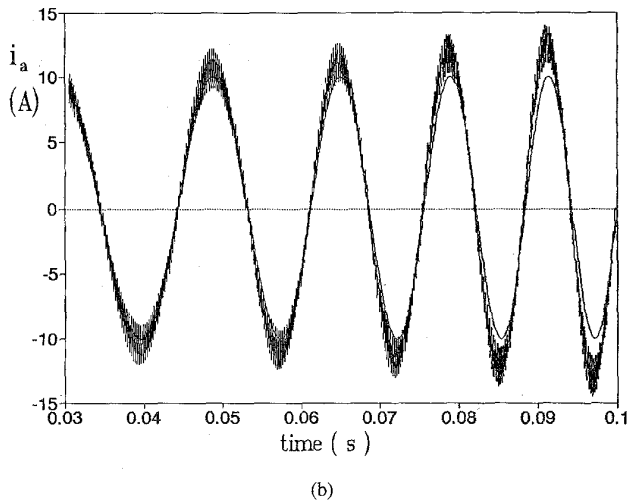
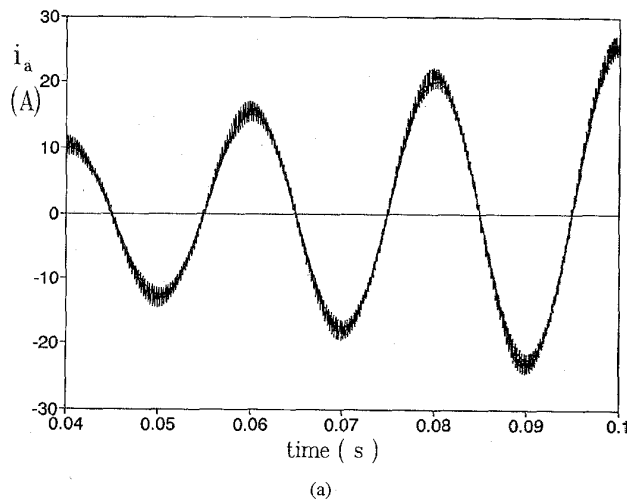


Fig. 14. (a) Response of phase A current to linear increase in reference amplitude (modified controller). (b) Response of phase A current to linear increase in reference frequency.

The response of the modified controller to a step change in reference is shown in Fig. 13. The large spike is again caused by the current prediction algorithm. But the response can be seen to be very fast, with a small settling time. Fig. 14 shows the responses of phase A current to a linear increase in reference amplitude and a linear increase in frequency at a rate of 0.325 Hz per sampling period. The increase in current error as the frequency increases is evident.

Fig. 15 shows phase A current when the inverter operates in saturation. The phase emf amplitude for this case is 250 V. Note that the maximum amplitude of the inverter line-to-neutral voltage is $(2/3)E \cos(30^\circ) = 260$ V in steady state for linear operation, so that with the above emf amplitude the inverter operates in saturation. The voltage reference V^* is outside the hexagonal region of synthesizable vectors for certain intervals of time. This nonlinear operation of the inverter causes the load current to be distorted with low-order harmonics.

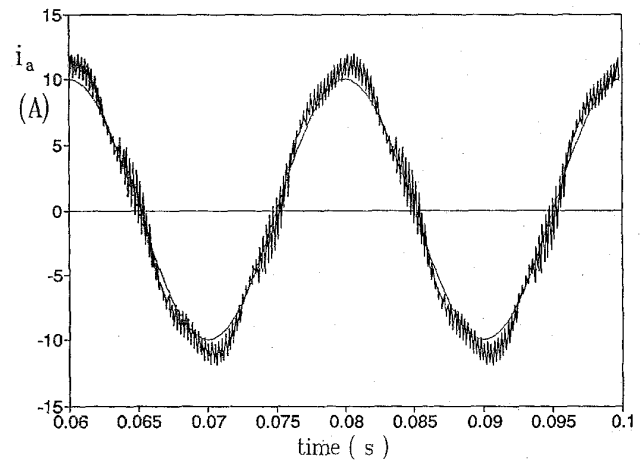


Fig. 15. Response of phase A current when the inverter operates in saturation (modified controller).

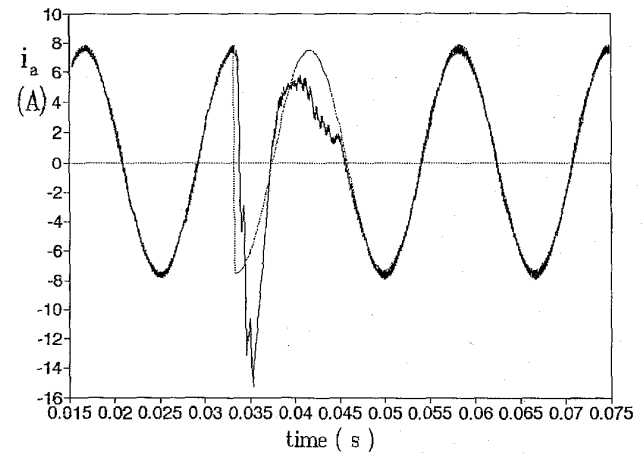


Fig. 16. Response of phase A current when transition from inversion to regeneration occurs (modified controller).

Fig. 16 shows response of phase A current when a sudden transition from inversion to regeneration (rectifier operation) occurs. This response is computed under similar operating conditions given in [9] and with the same parameters values. The computed current in the steady state appears to have less harmonic content than the measured current in [9]. The reason may be the better performance of space vector modulation at higher modulation levels than symmetrical sampled sine-triangle modulation [12]. However, it must also be kept in mind that nonideal effects in the experimental system in [9] contribute to the increase in harmonics in the measured current.

The transition in Fig. 16 is simulated by phase reversal of the current reference vector. The predicted current reference magnitude here is limited to 60 A. Without this limitation, the current spike caused by the step change in reference becomes very large. The response can be seen to be very fast with the current settling to steady state in less than one cycle. The variations in current amplitude after the transition in [9] are most probably caused by the dc voltage regulation

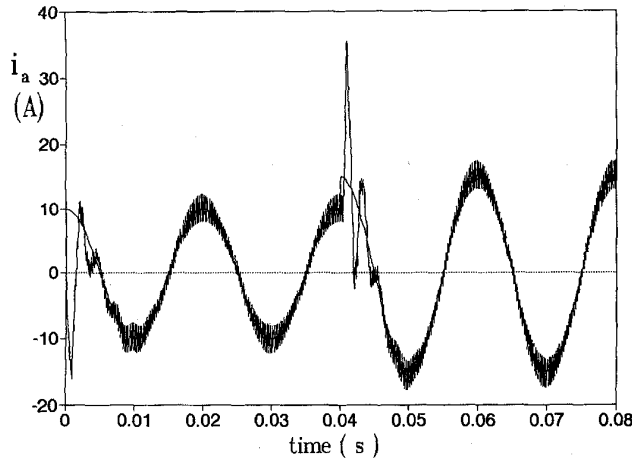


Fig. 17. Response of phase A current when load parameters deviate from nominal values.

loop, whereas the current controller here has no dc voltage regulation.

IX. ROBUSTNESS

When load parameters deviate from their nominal values, the poles of the control system shift away from their original locations given in (45). Analytical determination of the parameter value ranges in which the poles stay within the unit circle is quite involved, as the order of the system is high. Simulation studies have shown that both the predictive and the modified controllers are not very robust. The system is found to be not very sensitive to resistance value changes but becomes unstable when the inductance L deviates from its nominal value by more than $\pm 7\%$. It is also found that the emf prediction in (41) limits to a great extent the robustness of the system. When the predicted emf is replaced by the actual load emf in the simulations, the system remains stable for much wider ranges of the parameters. An alternative two-step-ahead prediction law for the emf is the following sinusoidal extrapolation

$$E_p(k) = e^{j2\omega T} E_e(k-2) \quad (58)$$

which gives zero prediction error in the steady state. The angular frequency ω can be extracted from measured current reference vectors as follows:

$$\begin{aligned} \omega T &\simeq \sin \omega T \\ &= \frac{1}{I_0 I_1} [I_d^*(k-1) I_q^*(k) - I_d^*(k) I_q^*(k-1)] \end{aligned} \quad (59)$$

where

$$I_0 = |I^*(k-1)|, \quad I_1 = |I^*(k)|.$$

With the prediction law in (58) simulations show that robustness is greatly improved with the system being stable for all $L \geq 3.1$ mH. This improvement is achieved at the expense of increased computation requirement. Fig. 17 shows the response of current with the modified controller ($N = 2$) and the emf prediction in (58), for $R = 0.5 \Omega$ and $L = 3.5$ mH. The response can be observed to be more oscillatory, meaning that the poles are now complex.

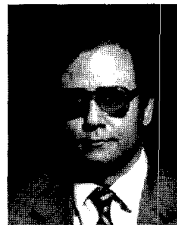
X. CONCLUSION

This paper has dealt with the current control of voltage-fed PWM inverters. It is stressed that the classical predictive controller has steady-state error and poor dynamic response problems due to computing delays. A new control strategy is proposed to alleviate the delay problem. However, the new controller is seen to lead to large steady-state errors particularly at higher operating frequencies. This necessitates the use of high-order prediction laws for the current reference and load emf. On the other hand, the transient performance has been greatly improved compared with the original predictive controller with computing delay.

Another problem of the predictive and modified controllers is sensitivity to load parameter variations, mainly caused by a polynomial-type prediction for back-emf. Sinusoidal prediction of emf is found to greatly improve robustness of the current control system.

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